

硕士学位论文

基于低分辨率视频的手势运动方向检测 Detection of Gesture Motion Direction based on Low-Resolution Video

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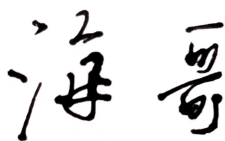
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摘 要

本文针对手机摄像头所获取的视频文件，进行手势运动方向的检测。针对低端摄像头视频图像的特点，本文采用了基于背景去除和肤色模型的方法对手部区域进行检测，并判别手部运动的方向。

首先，获取视频图像序列，即从视频文件中获取每一帧图像作为待检测的视频图像序列；其次，对获取的视频图像序列中的每一帧图像进行颜色模型转换、背景去除、图像二值化、形态学处理等预处理；然后，利用区域增长方法来检测视频图像序列中的手部连通区域，并计算每帧图像中手部区域的中心；最后根据图像序列中手部区域中心位置的变化来判断手部运动方向。

本文在 Visual c++6.0 开发环境下，借助于 OpenCV 开放平台，设计并实现了基于低端摄像头视频手势运动检测系统，得到了较好的检测效果。

关键词: 运动目标检测；颜色模型；区域增长

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Detection of Gesture Motion Direction based on Low-Resolution Video

Abstract

In this paper, the detection of hand motion direction is performed for the video files acquired by cell phone camera. Aiming at the characteristics of low-end camera video images, this paper adopts a method based on background removal and skin color model to detect the hand region and to discriminate the direction of hand motion.

Firstly, a video image sequence is acquired, i.e., each frame from the video file is acquired as a video image sequence to be detected; secondly, each frame in the acquired video image sequence is preprocessed with color model conversion, background removal, image binarization, morphological processing, etc.; then, a region growing method is used to detect the hand connectivity region in the video image sequence and to calculate the center of the hand region in each frame; finally, the hand region is detected based on the color model in the image sequence and the direction of hand movement is detected based on the color model. The center of the hand region; finally, the hand movement direction is judged based on the change of the center position of the hand region in the image sequence.

In this paper, under Visual c++6.0 development environment, with the help of OpenCV open platform, we designed and realized the hand gesture motion detection system based on low-end camera video, and got a better detection effect.

Key Words: Moving Target Detection; Color Model; Regional Growth

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